Introduction to Quantum Mechanics
Second Edition

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All errors, typographical and substantive, and other offenses, are entirely my own.

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Problem: 1.1

For the distribution of ages in Section 1.3.1:

(a) Compute $\langle j^2 \rangle$ and $\langle j \rangle^2$.

(b) Determine Δj for each j, and use Equation 1.11 to compute the standard deviation.

(c) Use your results in (a) and (b) to check Equation 1.12.

The data is

$$\begin{aligned} N(14) &= 1 \\ N(15) &= 1 \\ N(16) &= 3 \\ N(22) &= 2 \\ N(24) &= 2 \\ N(25) &= 5 \end{aligned}$$

(a) Our sample size is $N = 14$. Compute

$$\begin{aligned} \langle j^2 \rangle &= \sum_j j^2 P(j) \\ &= \frac{14^2 \cdot 1 + 15^2 \cdot 1 + 16^2 \cdot 3 + 22^2 \cdot 2 + 24^2 \cdot 2 + 25^2 \cdot 5}{14} \\ &= \frac{3217}{7} \approx 459.6 \\ \langle j \rangle^2 &= \left[\sum_j j P(j) \right]^2 \\ &= \left(\frac{14 \cdot 1 + 15 \cdot 1 + 16 \cdot 3 + 22 \cdot 2 + 24 \cdot 2 + 25 \cdot 5}{14} \right)^2 \\ &= \left(\frac{147}{7} \right)^2 = 21^2 = 441 \end{aligned}$$

Taking the difference produces the variance:

$$\sigma^2 = \langle j^2 \rangle - \langle j \rangle^2 = \frac{3217}{7} - 441 = \frac{130}{7} \approx 18.6$$

(b) Compute

$$\begin{aligned}\sigma^2 = \langle(\Delta j)^2\rangle &= \sum_j (\Delta j)^2 P(j) \\ &= \frac{(14-21)^2\cdot 1 + (15-21)^2\cdot 1 + (16-21)^2\cdot 3 + (22-21)^2\cdot 2}{14} \\ &\quad + \frac{(24-21)^2\cdot 2 + (25-21)^2\cdot 5}{14} \\ &= \frac{130}{7} \approx 18.6 \\ \sigma &= \sqrt{130/7} \approx 4.3\end{aligned}$$

(c) The computations confirm $\langle(\Delta j)^2\rangle = \langle j^2\rangle - \langle j\rangle^2$. Rooting gives us standard deviation $\sqrt{130/7} \approx 4.3$.

Problem: 1.2

(a) Find the standard deviation of the distribution in Example 1.1.

(b) What is the probability that a photograph, selected at random, would show a distance x more than one standard deviation away from the average?

(a) Using formula (1.12),

$$\begin{aligned}\langle x^2\rangle &= \int_0^h x^2 \frac{1}{2\sqrt{hx}}\, dx \\ &= \frac{1}{2\sqrt{h}}\left(\frac{2}{5}x^{5/2}\right)\Big|_0^h \\ &= \frac{h^2}{5}\end{aligned}$$

And since $\langle x\rangle = h/3$, we have $\langle x\rangle^2 = h^2/9$. Then compute

$$\sigma = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \sqrt{\frac{h^2}{5} - \frac{h^2}{9}} = \sqrt{\frac{4h^2}{45}} = \frac{2h}{3\sqrt{5}}$$

(b) We integrate the probability density over interval: $\left[h\left(1/3 - 2/3\sqrt{5}\right), h\left(1/3 + 2/3\sqrt{5}\right)\right]$[1] for the intermediate step:

$$\begin{aligned}\int_{h\left(1/3-2/3\sqrt{5}\right)}^{h\left(1/3+2/3\sqrt{5}\right)} \frac{1}{2\sqrt{hx}}\, dx &= \frac{1}{2\sqrt{h}}\left(2x^{1/2}\right)\Big|_{h\left(1/3-2/3\sqrt{5}\right)}^{h\left(1/3+2/3\sqrt{5}\right)} \\ &= \sqrt{\underbrace{\frac{1}{3} + \frac{2}{3\sqrt{5}}}_{A}} - \sqrt{\underbrace{\frac{1}{3} - \frac{2}{3\sqrt{5}}}_{B}}\end{aligned}$$

[1] The bounds fall within $[0, h]$, as $h\left(1/3 - 2/3\sqrt{5}\right) \approx 0.0352h$ and $h\left(1/3 + 2/3/\sqrt{5}\right) \approx 0.6315h$.

Observing that $\left(\sqrt{A}-\sqrt{B}\right)^2 = A + B - 2\sqrt{AB}$, compute and take the square root to simplify to

$$\frac{2}{3} - 2\sqrt{\frac{1}{9} - \frac{4}{45}} = \frac{2}{3} - \frac{2}{3\sqrt{5}} \quad \Longrightarrow \quad \sqrt{A} - \sqrt{B} = \sqrt{\frac{2}{3} - \frac{2}{3\sqrt{5}}} \approx 0.607$$

Lastly, subtracting from 1 yields a probability of ≈ 0.4.

Problem: 1.3

Consider the **gaussian** distribution

$$\rho(x) = Ae^{-\lambda(x-a)^2}$$

where A, a, and λ are positive real constants. (Look up any integrals you need.)

(a) Use Equation 1.16 to determine A.

(b) Find $\langle x \rangle$, $\langle x^2 \rangle$, and σ.

(c) Sketch the graph of $\rho(x)$.

(a) From Equation 1.16, we require

$$\int_{-\infty}^{\infty} Ae^{-\lambda(x-a)^2}\, dx = 1$$

For simplicity, let $u = x - a$. Then use the result

$$\int_{-\infty}^{\infty} e^{-\lambda x^2}\, dx = \sqrt{\frac{\pi}{\lambda}}$$

to deduce

$$\int_{-\infty}^{\infty} Ae^{-\lambda u^2}\, du = A\sqrt{\frac{\pi}{\lambda}} = 1 \quad \Longrightarrow \quad A = \sqrt{\frac{\lambda}{\pi}}$$

The intuition behind this substitution, which horizontally shifts $\rho(x)$ by a, is that the area underneath the Gaussian is independent of horizontal translation.

(b) To compute $\langle x \rangle$, make the same substitution $u = x - a$. Then $x = u + a$, and we compute

$$\langle x \rangle = \int_{-\infty}^{\infty} Axe^{-\lambda(x-a)^2}\, dx = A\left[\int_{-\infty}^{\infty} ue^{-\lambda u^2}\, du + \int_{-\infty}^{\infty} ae^{-\lambda u^2}\, du\right] = a$$

By antisymmetry of the first integrand about $x = a$, the first term evaluates to zero (put differently, odd in u). The second is simply the normalized Gaussian with a scale factor, leaving only a.

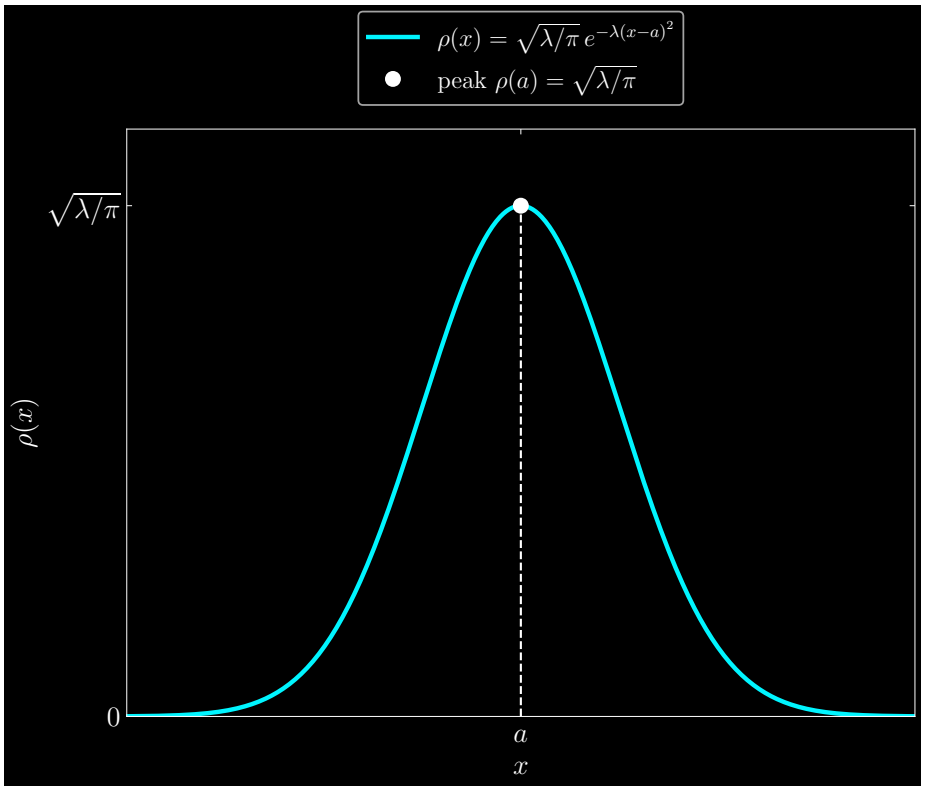
Next we have

$$\begin{aligned}\langle x^2\rangle &= \int_{-\infty}^{\infty} Ax^2 e^{-\lambda(x-a)^2}\,dx \\ &= A\left[\int_{-\infty}^{\infty} u^2 e^{-\lambda u^2}\,du + 2a\int_{-\infty}^{\infty} ue^{-\lambda u^2}\,du + \int_{-\infty}^{\infty} a^2 e^{-\lambda u^2}\,du\right] \\ &= \frac{1}{2\lambda} + a^2\end{aligned}$$

where we can conclude

$$\sigma = \sqrt{\langle x^2\rangle - \langle x\rangle^2} = \frac{1}{\sqrt{2\lambda}}$$

(c)



Problem: 1.4

At time $t = 0$ a particle is represented by the wave function

$$\Psi(x,0) = \begin{cases} A\dfrac{x}{a}, & \text{if } 0 \le x \le a, \\ A\dfrac{(b-x)}{(b-a)}, & \text{if } a \le x \le b, \\ 0, & \text{otherwise} \end{cases}$$

where A, a, and b are constants.

(a) Normalize Ψ (that is, find A, in terms of a and b).

(b) Sketch $\Psi(x,0)$, as a function of x.

(c) Where is the particle most likely to be found, at $t = 0$?

(d) What is the probability of finding the particle to the left of a? Check your result in the limiting cases $b = a$ and $b = 2a$.

(e) What is the expectation value of x?

(a) Normalizing the square-integrable wave function requires

$$A^2\left[\int_0^a \left|\frac{x}{a}\right|^2 dx + \int_a^b \left(\frac{b-x}{b-a}\right)^2 dx\right] = 1$$

We start by computing the integrals, which gets you

$$\begin{aligned} \int_0^a \left(\frac{x}{a}\right)^2 dx + \int_a^b \left(\frac{b-x}{b-a}\right)^2 dx &= \frac{x^3}{3a^2}\bigg|_0^a - \frac{(b-x)^3}{3(b-a)^2}\bigg|_a^b \\ &= \frac{a}{3} + \frac{b-a}{3} \\ &= \frac{b}{3} \end{aligned}$$

and leads to

$$A^2 = \frac{3}{b} \quad \Longrightarrow \quad A = \sqrt{\frac{3}{b}}$$

(b)